GERBANG LOGIKA

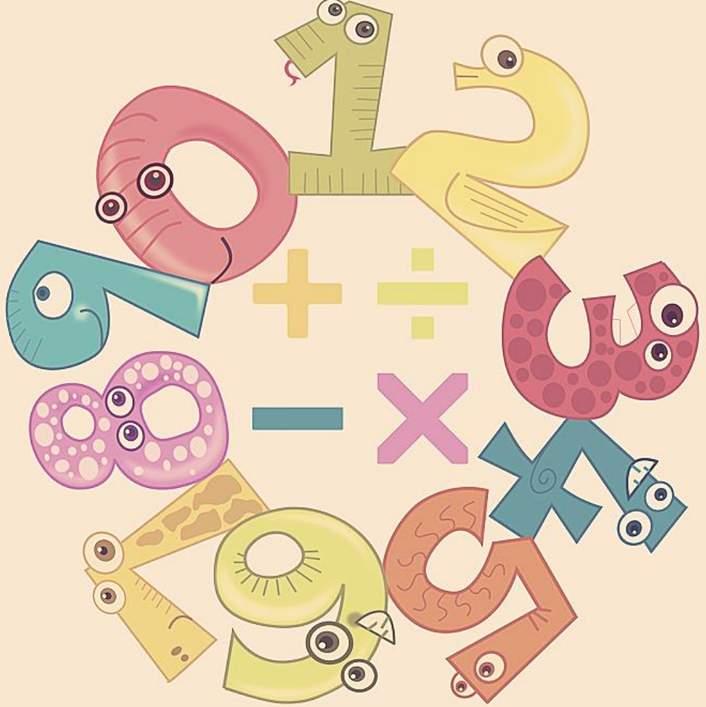
TEKNIK TELEKOMUNIKASI
POLITEKNIK NEGERI BANDUNG



MATERI

- OPERASI MATEMATIKA
- KONSEP DASAR RANGKAIAN LOGIKA
- TRUTH TABLE
- JENIS-JENIS GERBANG LOGIKA

OPERASI MATEMATIKA



ALJABAR LINEAR

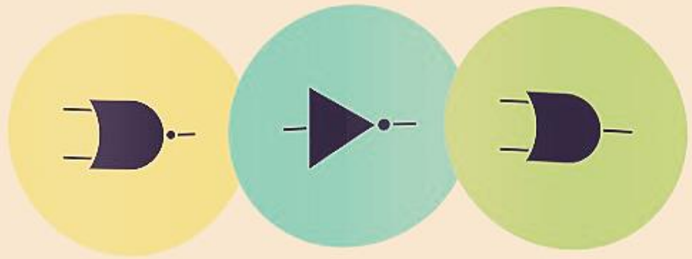
BASIS 10
DESIMAL
(0 - 9)

+ ; - ; x ; /

ALJABAR BOOLEAN

BASIS 2
BINER
(0 -1)

NOT, AND, OR, NOR, NAND, EX-OR, EX-NOR



KONSEP DASAR RANGKAIAN LOGIKA



Skema Input dan Output Gerbang Logika

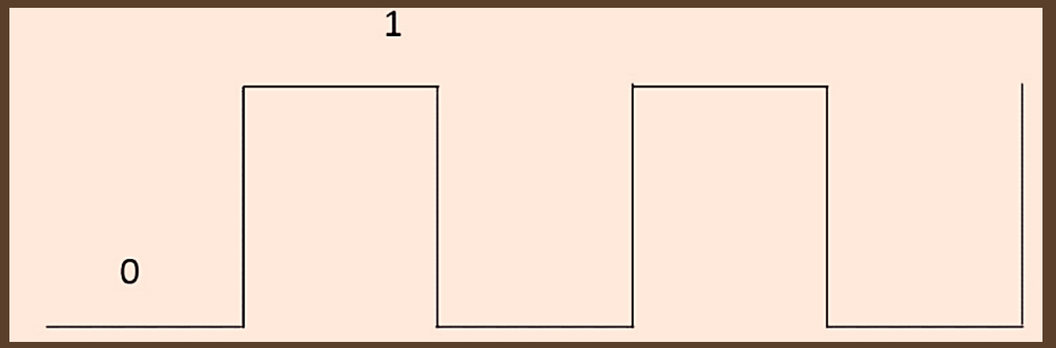


Diagram Logika 1 dan 0

TABEL KEBENARAN
TRUTH TABLE

Tabel kebenaran (*truth table*) merupakan tabel yang menunjukkan pengaruh pemberian level logika pada input suatu rangkaian logika terhadap keadaan level logika outputnya.

INPUT	OUTPUT
A	Y
0	
1	

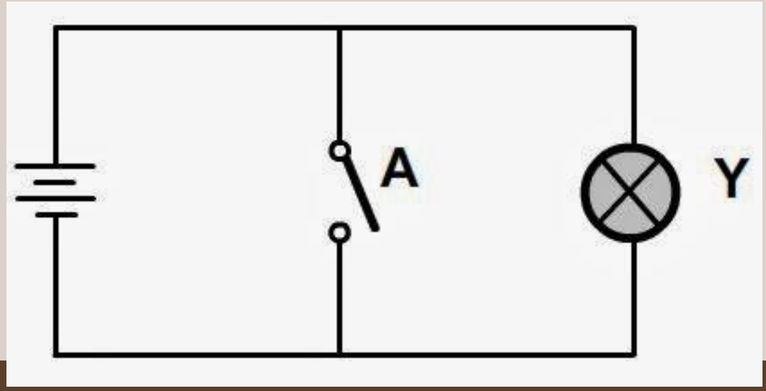
INPUT		OUTPUT
A	B	Y
0	0	
0	1	
1	0	
1	1	

INPUT			OUTPUT
A	B	C	Y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

OPERASI LOGIKA NOT (INVERTER)

IC TTL 7404

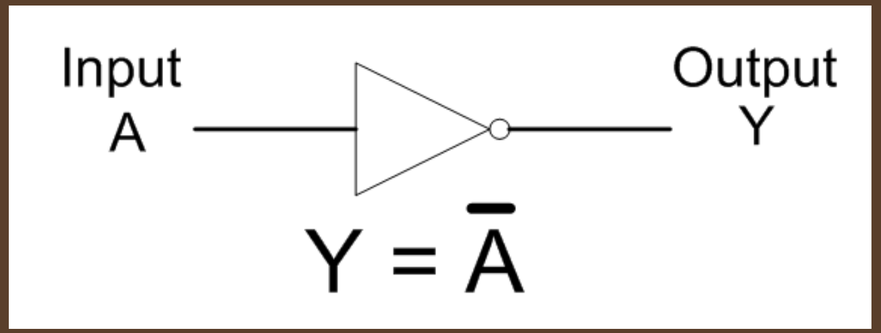
Fungsi yang membalik sebuah variabel Biner



Rangkaian Ekuivalen/setara

A	Y
0	1
1	0

Tabel Kebenaran Gerbang NOT



Simbol Rangkaian NOT

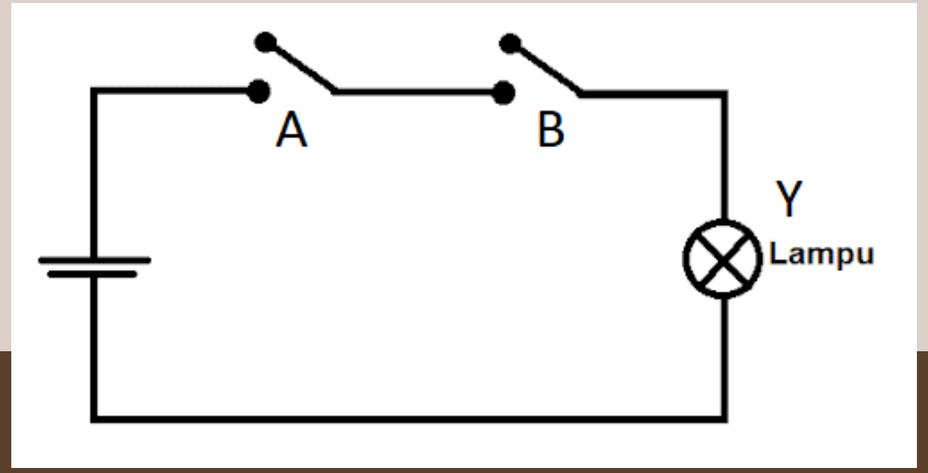
OPERASI LOGIKA

AND

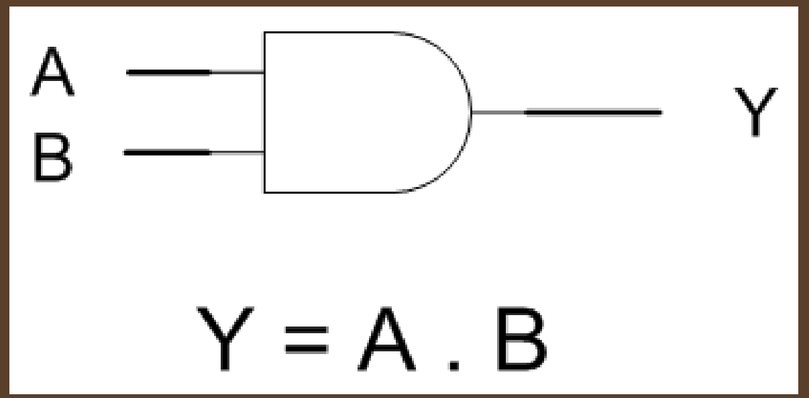
IC TTL 7408

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Tabel Kebenaran Gerbang AND



Rangkaian Ekuivalen/setara

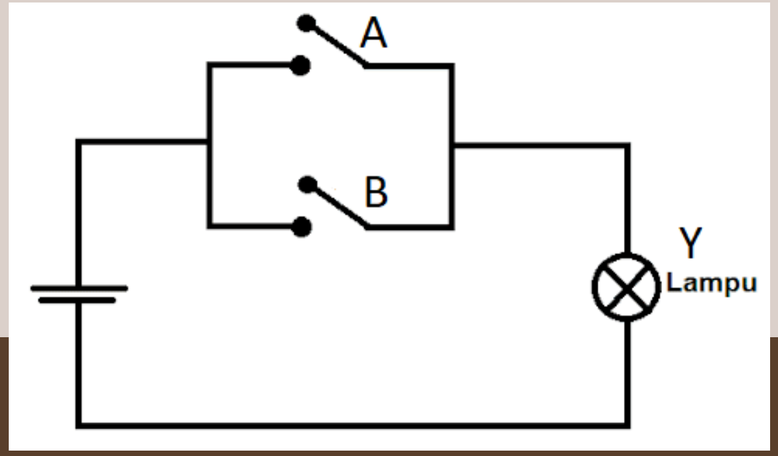


Simbol Rangkaian AND

OPERASI LOGIKA

OR

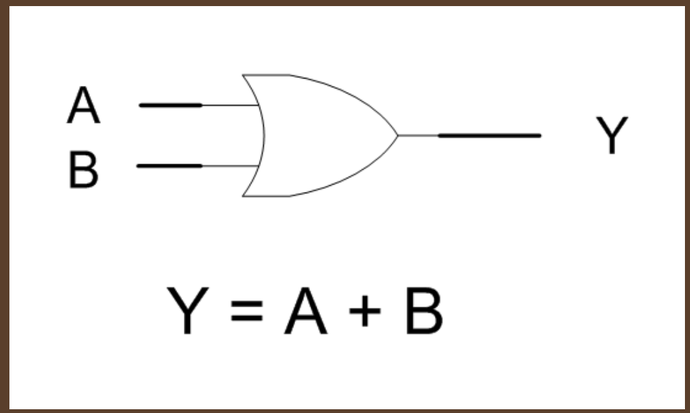
IC TTL 7432



Rangkaian Ekuivalen/setara

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Tabel Kebenaran Gerbang OR

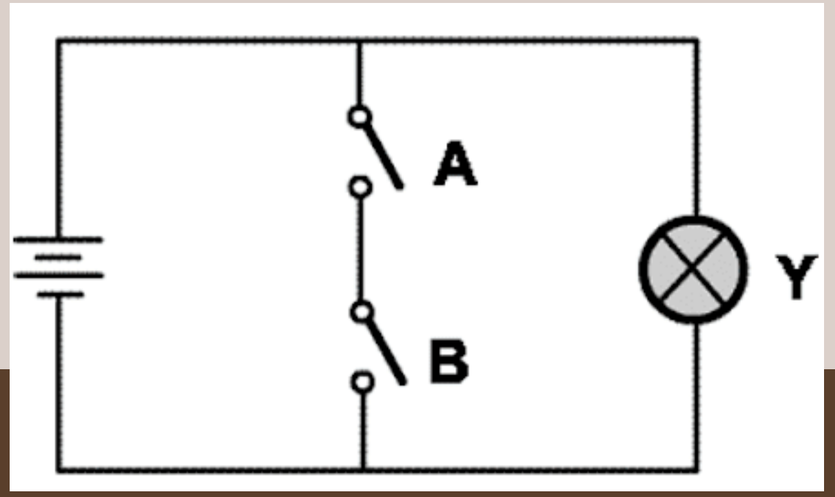


Simbol Rangkaian OR

OPERASI LOGIKA

NAND

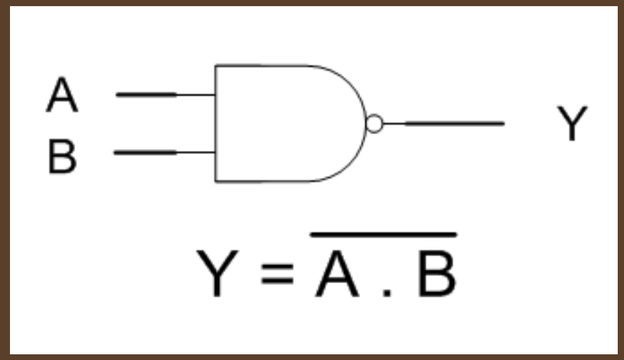
IC TTL 7400



Rangkaian Ekuivalen/setara

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Tabel Kebenaran Gerbang NAND



Simbol Rangkaian NAND

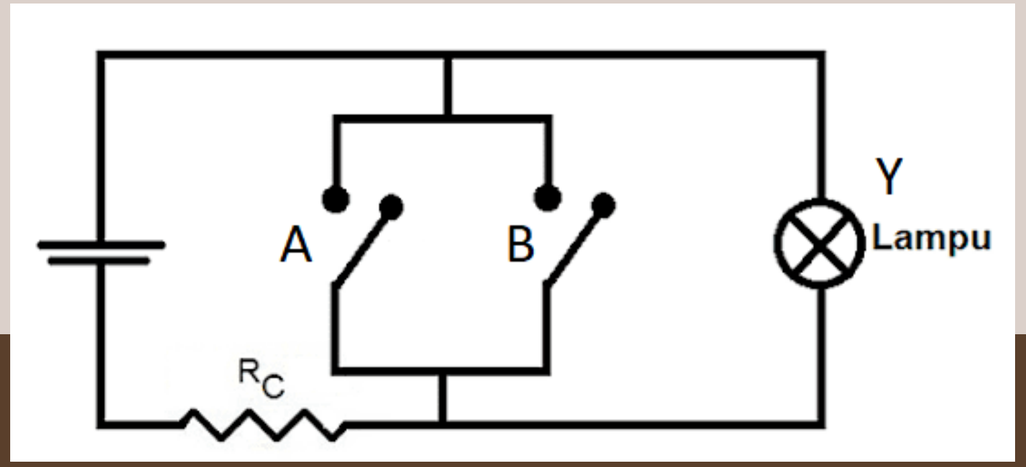
OPERASI LOGIKA

NOR

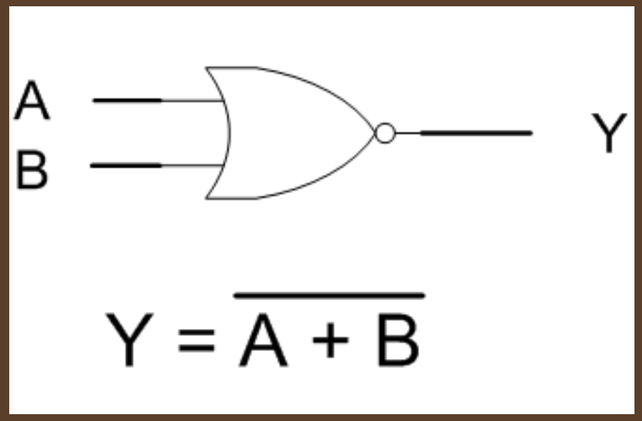
IC TTL 7402

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Tabel Kebenaran Gerbang NOR



Rangkaian Ekuivalen/setara



Simbol Rangkaian NOR

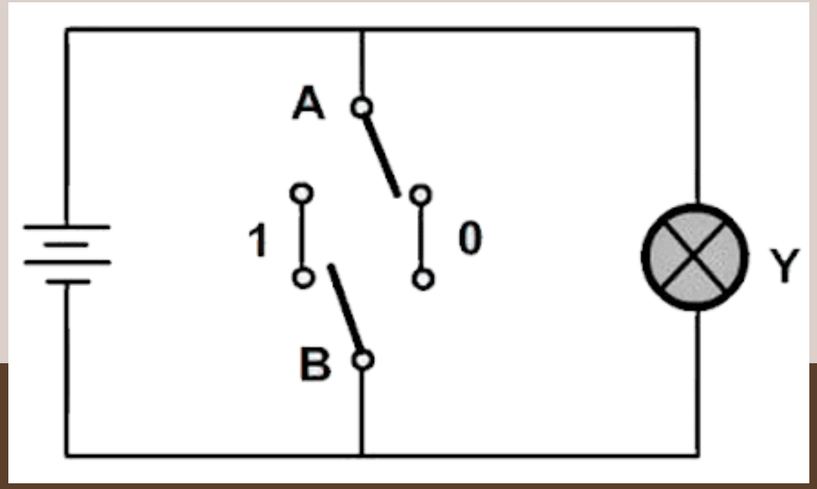
OPERASI LOGIKA

X-OR

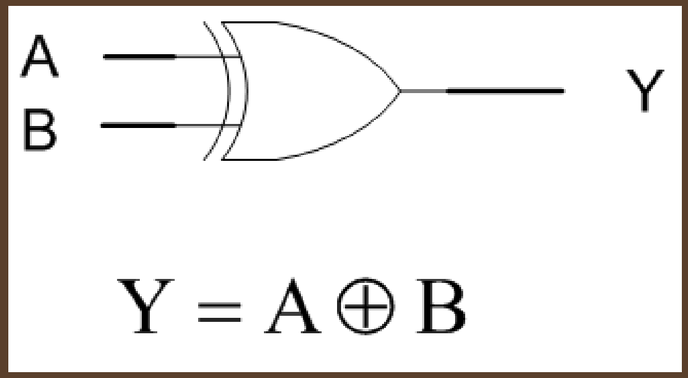
IC TTL 7486

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Tabel Kebenaran Gerbang X-OR

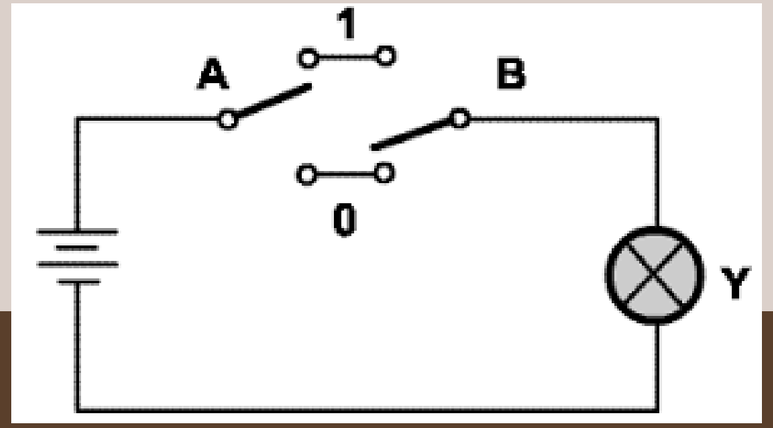


Rangkaian Ekuivalen/setara



Simbol Rangkaian X-OR

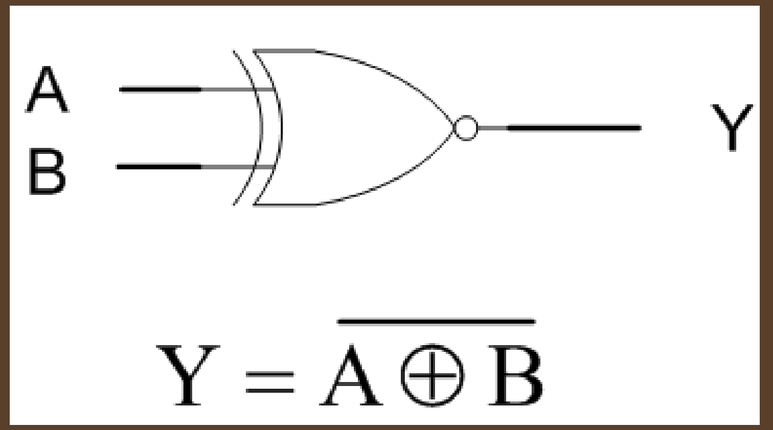
OPERASI LOGIKA
X-NOR



Rangkaian Ekuivalen/setara

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Tabel Kebenaran Gerbang X-NOR



Simbol Rangkaian X-NOR



Thank you!